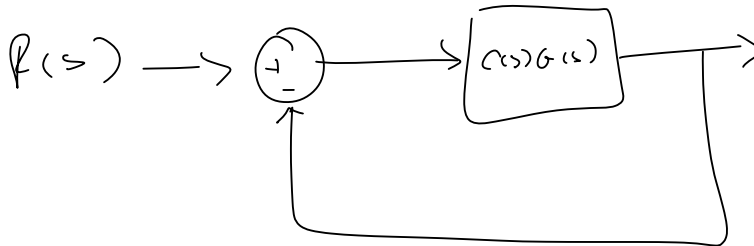


Assignment 14 (ELEC 341 L14_RLLeadLag)

Problem 1:

For a control system with open loop transfer function of $C(s).G(s) = C(s)/[s(s+4)]$, design a lead compensator by selecting $C_{lead}(s)$ so that the damping frequency is 4 rad/s and $\zeta = 0.707$.

$$G(s) = \frac{1}{s(s+4)}$$



$$\omega_d = 4 \frac{\text{rad}}{\text{s}}, \quad \zeta = \frac{1}{\sqrt{2}}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \quad \omega_n = 4\sqrt{2}$$

$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}} \rightarrow \frac{1}{\sqrt{2}}$$

$$s_d = -\zeta\omega_n \pm j\omega_d$$

$$C_{lead}(s) = K \frac{s + z_{lead}}{s + p_{lead}}$$

$$\text{let } z_{lead} = |p_c(s)|$$

$$z_{lead} = 2\sqrt{2}$$

$$s_d = -4 + j4$$

$$\angle \frac{s + z_{lead}}{s + p_{lead}} = \phi, \quad \phi = 180^\circ - \angle(s_d)$$

$$\phi = 180^\circ - 135^\circ$$

$$\phi = 45^\circ$$

$$G(s_d) = \frac{1}{(-4 + j4)(-4 + j4)} = \frac{1}{(-4 + j4)(j4)} = \frac{1}{-j16 - 16}$$

$$\angle G(s_d) = 180^\circ - \tan^{-1}(1)$$

$$\angle G(s_d) = 135^\circ$$

$$\angle \frac{s + z_{lead}}{s + p_{lead}} = \phi$$

$$\angle \frac{s_d + z_{lead}}{s_d + p_{lead}} = \phi$$

$$s + z_{lead} = f_m(s_d)$$

$$s_d = \operatorname{Re}(s_d) + f_m(s_d)$$

$$z_{lead} = -\operatorname{Re}(s_d)$$

$$\angle \frac{\operatorname{Im}(s_d)}{\operatorname{Re}(s_d) + \operatorname{Im}(s_d) + p_{lead}} = \phi$$

$$\operatorname{Re}(s_d) + \operatorname{Im}(s_d) + p_{lead}$$

$$90^\circ - \tan^{-1} \left(\frac{\operatorname{Im}(s_d)}{\operatorname{Re}(s_d) + p_{lead}} \right) = \phi$$

$$\tan(90^\circ - \phi) = \frac{\operatorname{Im}(s_d)}{\operatorname{Re}(s_d) + p_{lead}}$$

$$p_{lead} \tan(90^\circ - \phi) + \operatorname{Re}(s_d) \tan(90^\circ - \phi) = -\operatorname{Im}(s_d)$$

$$p_{lead} = \frac{-\operatorname{Im}(s_d) - \operatorname{Re}(s_d) \tan(90^\circ - \phi)}{\tan(90^\circ - \phi)}$$

$$p_{lead} = \frac{\operatorname{Im}(s_d) - \operatorname{Re}(s_d)}{\tan(90^\circ - \phi)}$$

$$= \frac{4}{\tan(45^\circ)} - (-4)$$

$$\boxed{p_{lead} = 8}$$

$$C_{lead}(s) = K \frac{s+4}{s+8}$$

$$K = \frac{1}{|L(s_d)|} \quad \text{where} \quad L(s) = C_{lead}(s)G(s) \quad s_d = -4 + j4$$

$$\boxed{K = 32}$$

$$L(s) = \frac{(s+4)}{s+8} \cdot \frac{1}{s(s+4)}$$

$$L(s) = \frac{1}{s(s+8)}$$

$$L(s_d) = \frac{1}{(-4+j4)(-4+j4+8)} \quad \begin{array}{l} (j4+4)(j4-4) \\ j^2 16 - 16j + 16j - 16 \\ -32 \end{array}$$

$$= \frac{1}{(-4+j4)(4+j4)} = \frac{1}{-32}$$

Thus,

$$\boxed{C_{lead}(s) = 32 \frac{s+4}{s+8}}$$